

Automated Geoprocessing for Infrastructure-Coupled Pluvial Flood Risk Assessment

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Abstract Heavy rainfall events pose growing hydrological challenges for urban areas, and capturing these challenges spatially presents existing analytical methods with a fundamental trade-off: physical-numerical models provide precise results but are too computationally intensive for large-scale applications on standard hardware, while topography-based GIS approaches are efficient but physically oversimplified. A Python-based, modular geoprocessing pipeline is used that combines both approaches and thereby specifically overcomes their respective limitations. It processes high-resolution digital elevation models (1×1 m), impervious surface data, and georeferenced channel infrastructure to generate comprehensive heavy rainfall hazard maps. A *Tiled-Buffered-Synchronized* architecture enables memory-intensive hydrological calculations even for large study areas on standard hardware. A *region-agnostic* design ensures the spatial transferability of the methodology. The pipeline is evaluated using the case study of Mierendorff Island (3.8 km^2) in the western centre of Berlin and creates hazard maps as well as identified flooding hotspots for four precipitation scenarios. This work presents an initial pipeline implementation using established GIS tools (SAGA GIS) for regional-scale applicability. While demonstrating proof-of-concept functionality, future work should adopt streaming-capable libraries for operational deployment.

Keywords Pluvial flood risk assessment · Accumulation based modeling · Urban stormwater runoff

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1 Introduction

As a result of climate change, heavy rainfall events are becoming increasingly frequent, which can lead to flash floods. A flash flood is defined as a sudden, locally confined flood with a high potential for damage [9]. Especially in urban areas, this can lead to sewer system overload and infrastructure damage and create surface waters [9]. Statistical analyses show that, with rising air temperatures and an increased capacity of the atmosphere to hold water vapor, an increase in the frequency and intensity of convective heavy precipitation is to be expected [9]. The German Weather Service (DWD) attributes this to human-induced climate change and refers, among other things, to the 2021 IPCC report, according to which heavy precipitation will become more frequent and intense on all continents as global warming increases [11].

The aim of this work is to assess and map these increasing risks and visualize potential hazards, based on

- a Digital Elevation Model (DEM),
- local weather and precipitation data,
- high-resolution impervious surface data, and
- information on sewer systems and water inlets

This enables the indication of regions with higher risk during heavy rainfall events. The output consists of maps that depict potential hazards to humans in the form of risk classes for various scenarios. The central design goal is a *region-agnostic* approach, i.e., a methodology that functions independently of the area under investigation. In addition, a modular pipeline structure was chosen so that individual steps and calculations can be processed and optimized independently of one another.

The development of the pipeline was guided throughout by the following research questions:

- RQ1: Are simple hydrological tools sufficient to provide an initial indication of heavy rainfall risk?
- RQ2: What topic-specific methods already exist, and how can they be effectively applied?
- RQ3: How can even large-scale areas be analyzed using only standard hardware?

First, the methodology, including the development and structure of the pipeline, is outlined, before a real-world analysis is presented using a case study.

2 Methodology

2.1 State of the Art

Hydrological studies and simulations can generally be conducted in two methodologically distinct ways: physical-numerical models and topography-based GIS tools. Both

pursue the same goal but differ significantly in their underlying tools, strengths, and limitations.

2.1.1 Physical-numerical models

This category includes fully dynamic coupled 1D/2D hydrodynamic models such as SWMM or MIKE21.

The *Storm Water Management Model* (SWMM) of the U.S. Environmental Protection Agency is a dynamic stormwater runoff model for single events and long-term simulations, specifically designed for urban areas. It divides the study area into sub-catchments and calculates the runoff paths within them, capturing both quantitative parameters (discharge, water depth) and qualitative ones (water quality). The model was released in 1971 and has been continuously developed since then [14]. *MIKE21* from the Danish Hydraulic Institute (DHI) is a widely used hydrodynamic simulation tool based on depth-averaged shallow water equations, enabling two-dimensional, time-continuous flow simulations [6].

The main drawback of both approaches is the high computational cost: A study published in April 2025 required 72 hours of computing time on a high-performance data center computer for a 3 km² study area in Aachen [15]. A high-resolution citywide analysis of a metropolis such as Berlin is therefore not feasible without extraordinary hardware resources.

2.1.2 Topography-based GIS approaches

The topography-based approach circumvents the problem of computational intensity by relying exclusively on GIS methods. Seleem et al. [16] investigated two such approaches for assessing the risk of stormwater flooding in urban environments in 2021.

In the *Fill-Spill-Merge* (FSM) method [3], depressions in the DEM are identified using the Priority Flood algorithm [2] and classified into a hierarchical *Depression Hierarchy*. For each grid cell, a flow direction to one of the eight neighboring pixels is determined using a Flowdirs matrix (D8 method). The FSM process proceeds in three steps:

- Precipitation is accumulated cell by cell and directed to the nearest depression (Fill)
- If this is full, the excess flows into the neighboring depression (Spill)
- If both are full, they merge into a higher-level meta-depression (Merge)

Since water redistribution in the Merge step occurs exclusively at the topological level of the hierarchical structure, the method is computationally efficient despite potentially large catchment areas [3]. The approach provides rapid results on water depth and places low demands on the input data, but does not allow for conclusions regarding flow velocities, discharge rates, or the degree of imperviousness [16].

The *Topographic Wetness Index* (TWI) [4] calculates a dimensionless hazard index from the DEM:

$$TWI = \ln \left(\frac{a}{\tan \beta} \right)$$

Here, a represents the upstream-contributing catchment area (in m^2 per unit of width) and β represents the local slope gradient. The larger the catchment area and the gentler the slope, the higher the TWI and the greater the flood risk associated with a pixel. The computational effort is minimal, as only a DEM is required. However, calibrating the classification thresholds necessitates a 2D reference model. This ensures that the approach is not transferable to other study areas, as threshold values are area-dependent [16].

Both topography-based approaches are significantly less computationally intensive than physical-numerical models, but do not allow for the full consideration of all relevant physical factors.

2.2 The Pipeline

2.2.1 Conceptual Approach

The core idea is the targeted combination of both classes of methods: Topography-based steps handle computationally efficient preprocessing tasks such as filling depressions and accounting for inlet infrastructure, while physical-numerical calculations based on high-resolution elevation models capture factors such as surface roughness, flow velocity, and water depth. Together, this pipeline enables reliable conclusions about flow direction and velocity, runoff and infiltration, as well as water depth.

2.2.2 Data Requirements

The pipeline requires three spatial input datasets. A digital elevation model (DEM) with a resolution of $1 \times 1 \text{ m}$ is required as the terrain model, since only at this level of detail can hydrologically relevant urban structures such as curbs or building foundations be accurately represented. In Germany, corresponding ATKIS DEM products are provided by the state surveying offices. In addition, comprehensive impervious surface data of the same resolution is required, from which the runoff coefficient C is derived for each pixel (see surface parameterization). Finally, for infrastructure coupling, the geodetic positions of the street inlets are required as a geographic point dataset. Since specific hydraulic measurements of the inlets are not uniformly available, each inlet structure is assigned a uniform inlet capacity of $Q = 50 \text{ l/s}$, corresponding to the upper control value according to DIN EN 1433 [8], which represents a conservative upper limit of the drainage capacity. The simulation is performed for four stationary precipitation scenarios with a reference duration of $D = 30 \text{ min}$ (10, 30, 50, and 90 mm), based

on the return periods of the KOSTRA-DWD-2020 dataset [10]. The scenarios cover a range of regularly occurring events below the channel design limit up to hydrological extreme events.

2.2.3 Hydrological Conditioning of the Digital Terrain Model

A central preprocessing step in the pipeline is the hydrological conditioning of the DTM in the form of a depression filling according to Wang & Liu [17]. In raw DTMs, there are points where all neighboring cells are higher, so that water remains stagnant there mathematically and no flow direction can be determined. The depression filling process iteratively raises such points until continuous runoff is possible. This step is particularly relevant in urban areas, as infrastructure objects such as bridges or culverts create artificial barriers in the DEM. A distinction is made between true hydrological sinks (depressions) and artifacts (bridges, culverts, measurement errors). The pipeline uses the freely available SAGA GIS module *Fill Sinks XXL*, which is based on the approach by Wang & Liu.

2.2.4 Surface Parameterization

Since the flow behavior of precipitation depends significantly on surface conditions, a purely topographic analysis is insufficient: On impervious surfaces such as asphalt, runoff behaves fundamentally differently than on green areas. Surface parameterization in the combined approach takes two forms. First, each pixel is assigned a runoff coefficient C with $0,0 < C < 1,0$, which indicates the relative effective runoff area: Asphalt releases nearly all precipitation as surface runoff ($C = 1,0$), whereas green areas often release only a fraction ($C = 0,2$). The categories were taken from DIN 1986-100 [7]. As a second parameterization, the Manning roughness coefficient n is introduced, which describes the natural flow resistance of the surface and is used both in the calculation of the flow velocity v and, via the Manning inversion, in the calculation of the water depth h [5].

2.2.5 Peak Discharge Modeling (Rational Method & Manning Inversion)

Using the conditioned flow directions and surface parameters, the peak discharge Q is calculated using the *Rational Method* according to Mulvaney [13]:

$$Q = C \cdot i \cdot A$$

Here, C represents the runoff coefficient, i represents the precipitation intensity (in mm/h), and A represents the contributing catchment area (in m²). The area A is not determined as a flat rate, but rather on a pixel-by-pixel basis from the flow accumulation along the conditioned flow directions, weighted by the respective runoff coefficient C of the contributing cells. As a result, topography and the degree of imperviousness are

both incorporated into the runoff calculation. The Rational Method is one of the most frequently used formulas in engineering hydrology for peak discharge estimation [1].

From Q , the water depth h is then derived on a pixel-by-pixel basis via the *Manning inversion*. The basis is the Gauckler-Manning-Strickler equation [5]:

$$v = \frac{1}{n} \cdot R^{2/3} \cdot S^{1/2}$$

For a wide, shallow channel, an approximation holds: $R \approx h$. Using the continuity equation for the specific discharge $q = Q/w$ (discharge per unit width w), we obtain:

$$q = \frac{1}{n} \cdot h^{5/3} \cdot S^{1/2}$$

Through algebraic manipulation, h can be calculated directly:

$$h = \left(\frac{q \cdot n}{S^{1/2}} \right)^{3/5}$$

The slope S is taken from the conditioned DEM, and n comes from the surface parameterization. Roughness thus enters the analysis twice: A high n dampens the flow velocity and, for the same discharge, leads to a greater calculated water depth, which underscores the physical plausibility of the approach.

2.2.6 Grid-Based Infrastructure Coupling

A distinctive feature of the combined approach is the explicit integration of the wastewater infrastructure in the form of inlet manholes. Georeferenced point data on the position and drainage capacity of the inlets are converted into the grid format of the DEM through rasterization, whereby each affected grid cell is assigned a drainage capacity Cp_{drain} (in m^3/s). If there are multiple inlets within a cell, the capacities are summed. The net surface runoff is then calculated as:

$$Q_{\text{net}} = \max(0, Q - Cp_{\text{drain}})$$

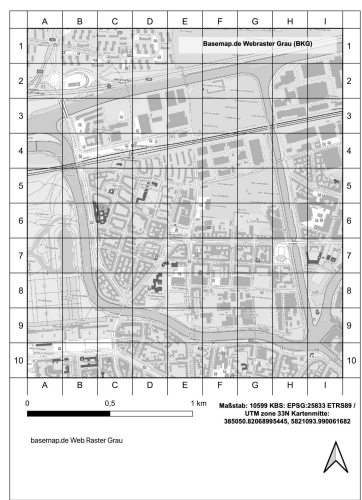
If the runoff is below the local absorption capacity, it is completely discharged into the sewer network. Otherwise, only the excess is propagated as surface runoff. If no manhole data is available, this step can be deactivated in accordance with the modular pipeline design without affecting the remaining calculation steps. The integration of the sewer infrastructure is essential for urban study areas: without it, no water would drain into the sewer system in heavily impervious areas, leading to a systematic overestimation of flood-prone areas.

2.2.7 Hazard Assessment

In the final step of the pipeline, the calculated values are converted into a qualitative risk classification. First, four depth classes are defined based on the flood volume. Since shallow but fast-flowing water can be more dangerous to humans than deep, stagnant water, the rating takes into account both water depth and flow velocity.

3 Results

3.1 Case Study: Berlin's Mierendorff Island



Mierendorff Island in western Berlin was selected as the study area. The 3.8 km² area is enclosed in a ring-shaped configuration by the Spree River and the Westhafen Canal, thereby forming a topographically and hydrologically well-defined watershed. At the same time, it exhibits a high degree of soil sealing and thus represents the typical characteristics of a densely built-up inner-city Berlin neighborhood (Figure 1).

The input data used are summarized in Table 1.

Fig. 1 Base map of the study area

Table 1 Input data used for the study area Berlin-Mierendorff Island

Dataset	Source	Year	Resolution
DEM	ATKIS, SenSBW Berlin	2021	1 × 1 m
Imperviousness	Berlin Environmental Atlas	2021	1 × 1 m
Street inlets	Geoportal Berlin	2012	Point data
Precipitation	DWD Open Data	since 1950	Station data

3.2 Hydrological Conditioning

Figure 2 shows the result of depression filling according to Wang & Liu [17]. The grayscale values encode the elevation difference between the original and the conditioned DEM, areas with significant depression filling appear as light zones.

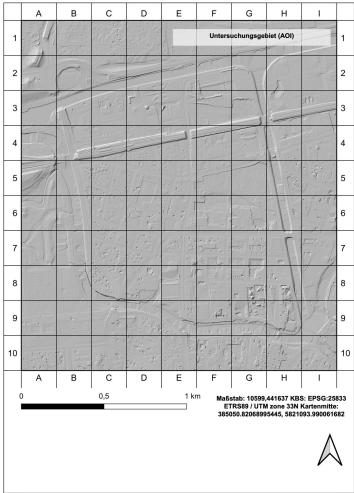


Fig. 2 Partial result of Depression filling

3.3 Water Depth Distribution

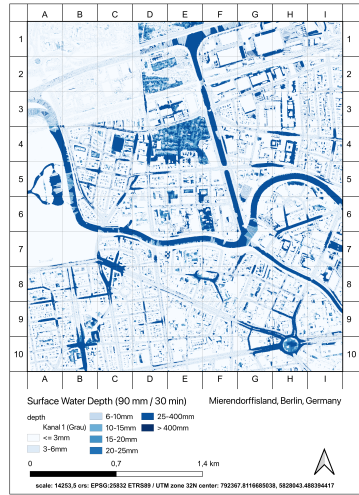


Fig. 3 Surface water depth for the 90 mm / 30 min scenario

Figure 3 shows the modelled surface water depth across the study area for a 90 mm / 30 min scenario. The building footprints derived from ALKIS (visible due to lack of flow in white) force flow routing around structures, while absolute depths remain modest (median 3 mm, 95th percentile 46 mm) due to the area’s high imperviousness, dense sewer coverage and specific pipeline parametrization. Yet flood-prone zones in low-lying streets, terrain depressions, and flow concentration along building edges are indicated.

3.4 Flow Velocity Distribution

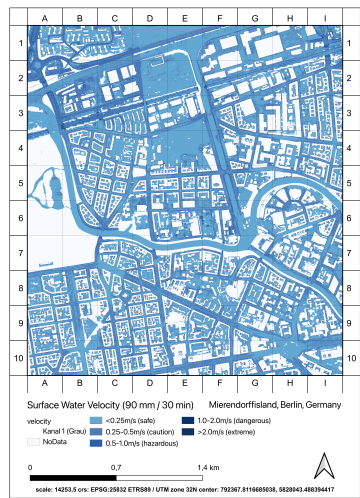


Fig. 4 Flow velocity for the 90 mm / 30 min scenario

3.5 Hotspot Identification

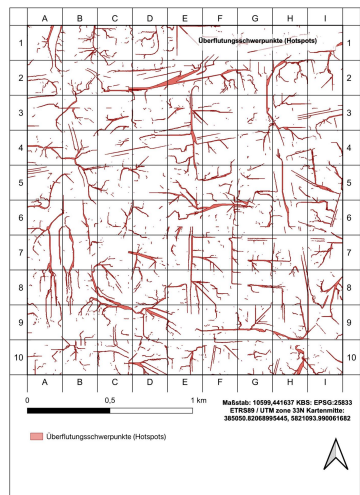


Fig. 5 Flood hotspots

Figure 4 shows the modelled surface flow velocity for a 90 mm / 30 min scenario. Narrow gaps between buildings concentrate flow and higher velocities are visible (dark blue). Lower velocities (light blue) occur along sheet-flow paths and in open areas. For the majority of the study area mean street-level remains below 1 m/s.

Figure 5 shows the flood hotspots identified in the study area based on a micro-watershed analysis. The areas marked in red exhibit a critical combination of high average imperviousness, high total precipitation volume, and high cumulative runoff accumulation and are therefore considered priority risk zones.

3.6 Hazard Rating

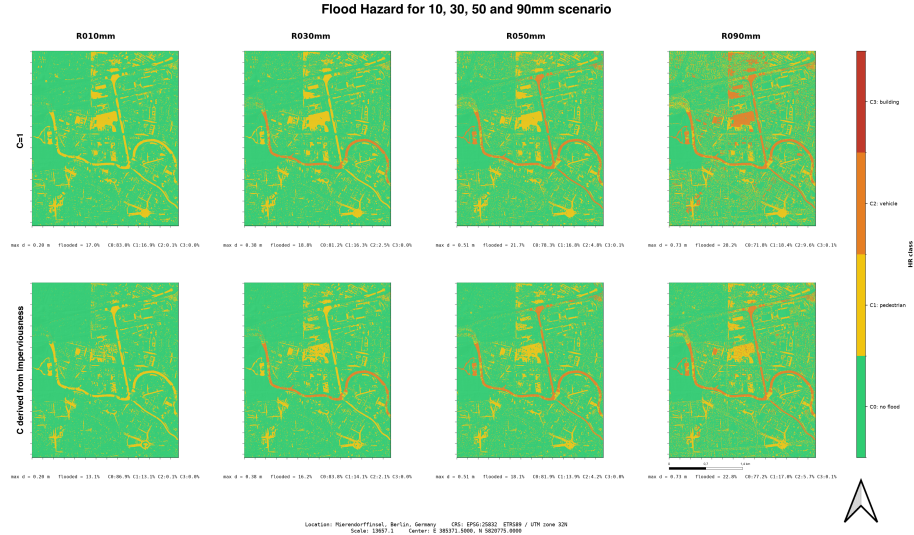


Fig. 6 Hazard Rating for all scenarios (A:10/B:30/C:50/D:90 mm/30min)

Figure 6 shows hazard for A:10, B:30, C:50 and D:90mm (30 minute duration) rainfall events. Derived from discharge ($q = d \cdot v$) creating a depth-dominant lookup matrix using the classes C0 (no standing water, $d < 5$ mm), C1 (pedestrian hazard, 5-30 mm), C2 (vehicle hazard, 30-100 mm), C3 (building hazard, >100 mm). Upper row shows full runoff scenario (runoff coefficient $C = 1$) the bottom row shows data with specific runoff coefficient derived from imperviousness map ($C = f(\text{surface cover})$).

4 Discussion

4.1 Methodological Value

The approaches presented in Section 2.1 represent the two ends of a methodological spectrum, between which the combined one has been specifically positioned. Although physical-numerical models such as SWMM or MIKE21 offer the most precise description of urban flooding processes, their computational demands fundamentally limit their applicability on a large scale: As already outlined in Section 2.1.1, a city-wide calculation at the required resolution is virtually impossible with current hardware capacity (RQ3). Topography-based approaches such as FSM or TWI solve the computational

problem but do not provide physically robust statements regarding flow velocities. Water volumes and volume balance, impervious surfaces, and sewer infrastructure are considered only indirectly, if at all [16]. Furthermore, TWI cannot be transferred to other study areas without a reference model, which fundamentally contradicts the pipeline’s *region-agnostic* design goal.

It aims to handle these limitations through two parameterization approaches: The discharge coefficient C according to DIN 1986-100 is incorporated into the flow accumulation with pixel-level accuracy, so that impervious surfaces directly affect the weighted catchment area. Furthermore, the Manning inversion allows for the calculation of water depth h and flow velocity v from the volumetric discharge, which represents a qualitative leap in the significance of the results compared to a purely static sink-filling analysis. As a distinguishing feature from both reference approaches, the combined approach also integrates georeferenced manhole data as a capacity grid directly into the runoff balance, thereby enabling quantifiable statements about where the sewer system significantly reduces damage and where its capacity is exceeded.

4.1.1 Answers to the Research Questions

The three research questions formulated in the introduction can be summarized as follows based on the development and application of the pipeline.

RQ1 (*Are simple hydrological tools sufficient for an initial hazard indication?*): In principle, yes. Flow accumulation and basin filling already outline plausible spatial patterns. However, additional physical parameters such as discharge coefficients, Manning’s roughness coefficient, and infrastructure capacity are required for robust and, above all, quantitatively interpretable statements regarding water depth and flow velocity.

RQ2 (*What topic-specific methods exist, and how can they be effectively applied?*): The Rational Method according to Mulvaney [13] and the sink-filling algorithm according to Wang & Liu [17] are established tools in engineering hydrology and hazard assessment. This project integrates these into a common, automated grid pipeline, thereby making them usable in combination for city-wide high-resolution analyses.

RQ3 (*How can large areas be analyzed on standard hardware?*): The *Tiled-Buffered-Synchronized* architecture provides an architectural solution to the out-of-memory problem of hydrological algorithms when dealing with high-resolution grids. Its application to the Mierendorff Island demonstrates that the concept works in practice and is scalable to larger study areas.

4.2 Limitations

The methodological strengths of the combined approach are inextricably linked to a series of simplifications that represent deliberate design choices in favor of computational efficiency and portability, but which limit the physical completeness of the model. These must be explicitly taken into account when interpreting the results.

4.2.1 Hydrological Simplifications

The key hydrological simplification lies in the use of the Rational Method as a steady-state peak discharge model. The method provides the maximum discharge Q_{surf} for a given precipitation scenario, but makes no statement about the temporal course of the discharge event. Time to peak, peak duration, and recession behavior cannot be determined using this approach. The results thus describe a stationary worst-case scenario, not the temporal dynamics of a real event. Closely related to this is the implicit assumption that all sub-areas of a catchment contribute to the peak discharge simultaneously. This assumption is reasonable for small, heavily impervious areas such as Mierendorff Island, but becomes increasingly physically implausible for larger study areas and leads to a systematic overestimation of peak discharge.

At the level of the flow direction algorithm, the pipeline uses the D8 method, in which each pixel deterministically discharges its water to the steepest of its eight neighboring pixels. In flat terrain, this results in unrealistically concentrated flow paths. Multiple-flow-direction algorithms (MFD), which distribute runoff proportionally to the gradient across several neighboring pixels, would produce a more physically realistic distribution. However, they could not be integrated into the selected SAGA-GIS workflow without significant additional effort.

4.2.2 Limitation regarding the runoff coefficient

During development, an implementation error was identified that limits the validity of key output variables from the pipeline and must be taken into account when interpreting the results.

As described in Section 2.2.4, the methodological concept involves deriving the runoff coefficient C with pixel-level accuracy from comprehensive impervious surface data and assigning each pixel an individual value according to DIN 1986-100 based on its surface characteristics. In the current implementation, however, C is not parameterized based on land cover type but is applied uniformly as a constant value across the entire study area. The spatially differentiated imperviousness information is thus not incorporated into the calculation.

This limitation affects several subsequent calculation steps. In peak discharge mod-

eling using the Rational Method ($Q = C \cdot i \cdot A$), the impervious-weighted catchment area is not determined correctly, as the weighting of the contributing cells by their respective surface type is omitted. Since Q in turn forms the basis of the Manning inversion, this error propagates into the calculated water depths h and flow velocities v . In highly heterogeneous urban areas with marked alternation between impervious and pervious surfaces, a systematic overestimation of peak discharge in less impervious areas is to be expected.

Correcting this error requires the correct pixel-by-pixel linking of the imperviousness classes with the corresponding C values in the accumulation step, followed by a re-application of the pipeline to the study area.

4.2.3 Simplifications in Infrastructure Coupling

The integration of the sewer infrastructure is implemented as a static capacity balance, not as a hydraulic network model. The uniformly assigned absorption capacity of $Q = 50 \text{ l/s}$ according to DIN EN 1433 [8] represents a conservative upper limit. The actual capacity of an inlet manhole depends heavily on local factors such as condition, degree of contamination, and flow geometry, and may deviate significantly from this flat-rate value. Furthermore, the sewer system in the pipeline remains a pure discharge capacity without feedback to the surface. Hydraulic effects such as backwater in overloaded collection pipes or the overflow of water from the sewer network (*surcharge overflows*) are not taken into account, although they are not negligible in reality, particularly in the extreme scenarios that are especially relevant in flood risk assessments.

4.2.4 Uncertainties in Parameterization

The Manning roughness coefficient n is derived with pixel-level accuracy from impervious surfaces and is based on literature values for surface types that have not been calibrated for specific regions. Since n is used in the calculation of both h and v and is applied to h with an exponent of 0.6, systematic misestimates of this parameter have a disproportionately large impact on both output variables.

While the tile-based architecture enables working with established GIS tools (SAGA GIS), it limits computational efficiency. Migration to streaming-capable libraries such as RichDem could reduce processing time significantly for large-scale applications[12].

4.2.5 Data-related limitations

The DEM used does not contain any information about underground passages, underpasses, or bridge structures. In the raw DEM, these create artificial barriers that, although topologically resolved by conditioning according to Wang & Liu [17], are not

physically correctly represented: The actual flow path runs *under* the bridge, while the modeled path runs *over* it. Furthermore, the street inlet data dates from 2012 and does not reflect structural changes to the infrastructure. Finally, since no calibration data in the form of water level measurements or documented flood extents is available, the calculated water depth values should be understood as a qualitative risk indication and not as a quantitative forecast of specific water levels.

4.3 Validation & Transferability

Since implementation errors were identified in the pipeline, validation and transferability can only be considered together: A complete quantitative validation would require reference data from real heavy rainfall events in the study area (such as water level records, documented flood extents, or damage reports), combined with a sufficiently error-free model whose outputs can be meaningfully compared with observational data. Neither ambition is fully met within the scope of this work, given the implementation error described in Section 4.2.2. External validation is therefore reserved for a revised version of the pipeline.

Based on the available results, however, the structural plausibility of the pipeline can be assessed: As expected, the calculated flow paths follow low-lying streets and depressions in the terrain, hotspot localization is concentrated in hydrologically plausible areas with a high degree of imperviousness, and the scenarios respond consistently to increasing precipitation intensities. The consistency tests of the pipeline (seam line checks and volume balancing) quantitatively demonstrate that the tiling strategy does not generate hydraulic artifacts at the tile boundaries. These indications point to a fundamentally functional model concept, though they do not replace validation based on empirical event data. The current state of work is thus a viable methodological concept, and the pipeline is technically operational, but targeted revisions are essential for further practical application.

Transferability is conceptually already achieved: The tiling architecture is region-independent, the standard values for discharge coefficients according to DIN 1986-100 and Manning roughness coefficients are not calibrated to Berlin. In practice, however, transferability depends on data availability, which cannot be assumed everywhere. A 1×1 m DEM is available nationwide in Germany through the state surveying offices, but is by no means standard internationally, and impervious surface data of comparable resolution and georeferenced inflow data vary greatly in availability and timeliness by municipality. The pipeline is thus methodologically transferable, but in terms of data still limited to comparatively well-mapped urban contexts. A sensible next step would be to apply the revised pipeline to a second study area with different topographic and infrastructural characteristics to empirically validate the *region-agnostic* claim.

5 Conclusion

This project was developed as an automated, Python-based geoprocessing pipeline that combines topography-based GIS methods with physical-numerical calculations, thereby conceptually bridging the methodological gap between computationally intensive hydrodynamic models and physically simplified GIS approaches. For any stationary precipitation scenario, it provides comprehensive hazard maps, imperviousness-weighted flow accumulation maps, flood hotspots, and delta maps to quantify the relief effect of the sewer infrastructure. Application to Berlin's Mierendorff Island (3.8 km²) shows that these outputs can be generated with 1 × 1 m resolution on standard hardware.

In its current state, the pipeline constitutes a proof of concept. The identified implementation error limits the interpretability of individual output variables and requires systematic correction. The integrated consistency checks and delta analyses have fulfilled their function as validation tools and made the limitations of the current state transparent. Furthermore, the stationary analysis without temporal discharge dynamics, the simplified infrastructure coupling without backwater modeling, and the lack of empirical calibration are deliberate design choices made in favor of computational efficiency and transferability. They limit interpretability accordingly. The calculated water depth values are reliable as a qualitative risk indicator, not as a quantitative forecast of specific water levels, which should be explicitly communicated whenever the pipeline is used.

As the next immediate step, the implementation errors must first be corrected and the pipeline reapplied to the study area. This would create a basis for external validation using real event data, for example by comparing calculated hotspots with documented damage reports from past heavy rainfall events in Berlin. In the medium term, applying the model to a second study area with different topographic and infrastructural characteristics would be an important step toward empirically substantiating the *region-agnostic* claim and demonstrating the extent to which data requirements limit transferability to other municipalities or international contexts. In the long term, the modular architecture offers concrete starting points for methodological extensions: replacing the D8 algorithm with an MFD approach, time-dynamic precipitation modeling using synthetic hydrographs, and a true 1D/2D coupling of sewer hydraulics to model backwater effects. The extent to which the latter could still be implemented at the city level using standard hardware remains an open question for future work.

References

- [1] Baiamonte G (2020) A rational runoff coefficient for a revisited rational formula. *Hydrological Sciences Journal* 65(1):112–126, DOI 10.1080/02626667.2019.1682150, URL <https://www.tandfonline.com/doi/full/10.1080/02626667.2019.1682150>

- [2] Barnes R, Lehman C, Mulla D (2014) Priority-flood: An optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Computers & Geosciences* 62:117–127, DOI 10.1016/j.cageo.2013.04.024, URL <https://linkinghub.elsevier.com/retrieve/pii/S0098300413001337>
- [3] Barnes R, Callaghan KL, Wickert AD (2021) Computing water flow through complex landscapes – Part 3: Fill–Spill–Merge: flow routing in depression hierarchies. *Earth Surface Dynamics* 9(1):105–121, DOI 10.5194/esurf-9-105-2021, URL <https://esurf.copernicus.org/articles/9/105/2021/>
- [4] Beven KJ, Kirkby MJ (1979) A physically based, variable contributing area model of basin hydrology / Un modèle à base physique de zone d’appel variable de l’hydrologie du bassin versant. *Hydrological Sciences Bulletin* 24(1):43–69, DOI 10.1080/02626667909491834, URL <http://www.tandfonline.com/doi/abs/10.1080/02626667909491834>
- [5] Chow VT (1959) Open-channel hydraulics
- [6] Danish Hydraulic Institute (2026) MIKE 21 Flow Model FM. URL https://manuals.mikepoweredbydhi.help/latest/Coast_and_Sea/MIKE_FM_HD_2D.pdf
- [7] DIN 1986-100 (2016) Abflussbeiwerte zur Berechnung des Niederschlagswasserabflusses
- [8] DIN EN 1433 (2005) Entwässerungsrinnen für Verkehrsflächen - Klassifizierung, Bau- und Prüfgrundsätze, Kennzeichnung und Beurteilung der Konformität
- [9] Fritsch U, Zebisch M, Voß M, Linsenmeier M, Kahlenborn W, Porst L, Hölscher L, Wolff A, Hardner U, Schwartz K, Wolf M, Schmuck A, Schönthaler K, Nilson E, Fischer H, Fleischer C (2021) Teilbericht 3: Risiken und Anpassung im Cluster Wasser [Partial Report 3: Risks and Adaptation in the Water Cluster]. Ressortforschungsplan 22/2021, Umweltbundesamt, Dessau-Roßlau, URL https://www.umweltbundesamt.de/system/files/medien/479/publikationen/kwra2021_teilbericht_3_cluster_wasser_bf_211027_0.pdf
- [10] Junghänel T, Ostermüller J, Deutschländer T (2023) Grids of return periods of heavy precipitation (design precipitation) over Germany (KOSTRA-DWD) Raster der Wiederkehrintervalle für Starkregen (Bemessungsniederschläge) in Deutschland (KOSTRA-DWD). DOI 10.5676/DWD/KOSTRA-DWD-2020, URL https://opendata.dwd.de/climate_environment/CDC/help/landing_pages/doi_landingpage_KOSTRA_DWD_2020-en.html
- [11] Kaspar F, Deutschländer T, Junghänel T, Lengfeld K, Ostermüller J, Palarz A, Rauthe M, Rustemeier E, Walawender E, Winterrath T, Ziese M (2025) Niederschlagstrends und Entwicklung der Starkniederschläge in Deutschland (Stand: 2025) [Precipitation trends and development of heavy precipitation in Germany (as of 2025)]. Faktenpapier, Deutscher Wetterdienst, Deutschland, URL https://dwd.de/DE/leistungen/faktenpapier_extremwetter/niederschlagstrend-entwicklung_2025.pdf?__blob=publicationFile&v=1
- [12] Leistert H (2025) AccRo - Model Code. DOI 10.5281/ZENODO.17153807, URL <https://zenodo.org/doi/10.5281/zenodo.17153807>, language: de

- [13] Mulvany T (1851) On the use of self-registering rain and flood gauges in making observations of the relations of rain fall and flood discharges in a given catchment. Transactions of the Institution of Civil Engineers of Ireland Vol. IV(pt. II):18–33
- [14] Rossmann L (2015) Storm Water Management Model User's Manual Version 5.1. URL https://www.epa.gov/sites/default/files/2019-02/documents/epaswmm5_1_manual_master_8-2-15.pdf
- [15] Sañudo E, García-Feal O, Hagen L, Cea L, Puertas J, Montalvo C, Alvarado-Vicencio R, Hofmann J (2025) IberSWMM+: A high-performance computing solver for 2D-1D pluvial flood modelling in urban environments. Journal of Hydrology 651:132603, DOI 10.1016/j.jhydrol.2024.132603, URL <https://linkinghub.elsevier.com/retrieve/pii/S0022169424019991>
- [16] Seleem O, Heistermann M, Bronstert A (2021) Efficient Hazard Assessment for Pluvial Floods in Urban Environments: A Benchmarking Case Study for the City of Berlin, Germany. Water 13(18):2476, DOI 10.3390/w13182476, URL <https://www.mdpi.com/2073-4441/13/18/2476>
- [17] Wang L, Liu H (2006) An efficient method for identifying and filling surface depressions in digital elevation models for hydrologic analysis and modelling. International Journal of Geographical Information Science 20(2):193–213, DOI 10.1080/13658810500433453, URL <http://www.tandfonline.com/doi/abs/10.1080/13658810500433453>